Quantification: 536-63x_Hyp_RD_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	14
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3234.5	3016	1369.81	px^2
major_axis_length_px	lacuna shape	119.681	112.891	44.2572	px
minor_axis_length_px	lacuna shape	42.5255	40.7908	10.2373	px
aspect_ratio	lacuna shape	2.8427	2.861	0.8647	unitless
eccentricity	lacuna shape	0.9203	0.9369	0.0433	unitless
solidity	lacuna shape	0.7542	0.8127	0.1363	unitless
roots_count	headline	13.8571	11	8.2542	count
ring_length_r30_px	headline	404.857	327	205.102	px
ring_length_r60_px	reported	820.429	762.5	388.144	px
owned_length_px	ownership-dependent	1124.17	1030.23	889.806	px
edge_count	ownership-dependent	54.7143	45	43.0678	count
mean_edge_length_px	ownership-dependent	21.4311	21.7276	4.9182	px
perimeter_px	normalised	376.313	319.706	185.731	px
ring_area_r30_px2	normalised	12033.6	11325.5	3577.1	px^2
ring_area_r60_px2	normalised	27270.2	26964.5	7249.2	px^2
in_frame_fraction_r30	normalised	0.9947	1	0.0198	unitless
in_frame_fraction_r60	normalised	0.9845	1	0.0528	unitless
ring_density_r30	normalised	0.0322151	0.0314993	0.00878791	px^-1
ring_density_r60	normalised	0.0288727	0.0284843	0.00808058	px^-1
roots_per_100px_perimeter	normalised	3.5744	3.7104	0.7329	per 100 px
ring_attached_length_r30_px	network v2	357.214	287.5	208.346	px
ring_attached_length_r60_px	network v2	632.143	569	371.057	px
ring_length_w_r30_px	network v2	457.282	362.264	231.717	px
ring_length_w_r60_px	network v2	932.299	865.488	439.175	px
sholl_crossings_r10	network v2	9.7143	8	4.6315	count
sholl_crossings_r20	network v2	10.6429	10	4.4134	count
sholl_crossings_r30	network v2	11.4286	10.5	4.6693	count
loop_count	loops, report only	3.0714	1.5	4.2147	count
loop_edge_share	loops, report only	0.1288	0.1288	0.1186	share
ring_width_mean_r30_px	width	7.583	7.3941	0.6612	px

Per field
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measure	value	units
analysed_area_px2	950894	px^2
total_skeleton_length_px	25529	px
canalicular_length_density_per_px	0.0268474	px^-1
junction_count	484	count
junction_density_per_px2	0.00050899	px^-2
lacuna_count	14	count

lacunae_per_px2	1.472e-05	px^-2
skeleton_component_count	366	count
mean_component_length_px	69.7514	px
median_component_length_px	24	px
field_length_density_w_per_px	0.0304832	px^-1
field_density_without_flagged_per_px	0.025464	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	331	px
unattached_component_count	19	count
canalicular_length_density_attached_per_px	0.0268474	px^-1
field_length_density_with_unattached_per_px	0.0271955	px^-1
excluded_area_px2	52399	px^2
analysed_field_area_px2	996177	px^2
loop_count	51	count
loop_edge_share	0.1458	share
edge_count	1331	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7047	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25516	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5184	244.559	46.0658	5.3089	0.9821	0.5932	-0.3315	180.801	650.935	False
2	5306	151.268	65.5589	2.3074	0.9012	0.6017	0.1738	176.169	814.499	False
3	2838	120.116	56.4992	2.126	0.8825	0.5006	-0.0922	183.402	482.588	False
4	5032	145.049	50.087	2.8959	0.9385	0.6975	-0.3567	389.803	259.63	False
5	3194	132.785	39.2534	3.3827	0.9553	0.719	-0.3063	465.147	830.34	False
6	4023	130.997	44.5634	2.9396	0.9404	0.8529	-0.2947	478.273	373.626	False
7	2063	91.1753	31.0575	2.9357	0.9402	0.8892	-0.3885	511.078	505.802	False
8	2736	102.242	36.178	2.8261	0.9353	0.8653	-0.4465	632.235	931.07	False
9	2738	96.5558	37.4594	2.5776	0.9217	0.8784	-0.3793	679.917	399.446	False
10	3288	105.667	42.059	2.5123	0.9174	0.8761	-0.3815	724.683	177.336	False
11	1381	83.2261	24.5172	3.3946	0.9556	0.81	-0.4203	774.605	822.052	False
12	1183	68.5981	39.5227	1.7357	0.8173	0.5796	-0.4587	845.742	872.086	False
13	4334	128.945	44.2583	2.9135	0.9393	0.8796	-0.1396	903.815	374.562	False
14	1983	74.3475	38.2775	1.9423	0.8573	0.8154	-0.0273	970.974	775.836	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	35	744	1400	1505.51	92	16.3642	7.5136
2	25	698	1290	2619.3	116	22.5802	7.4044
3	19	723	1433	1324.53	81	16.3522	6.6187

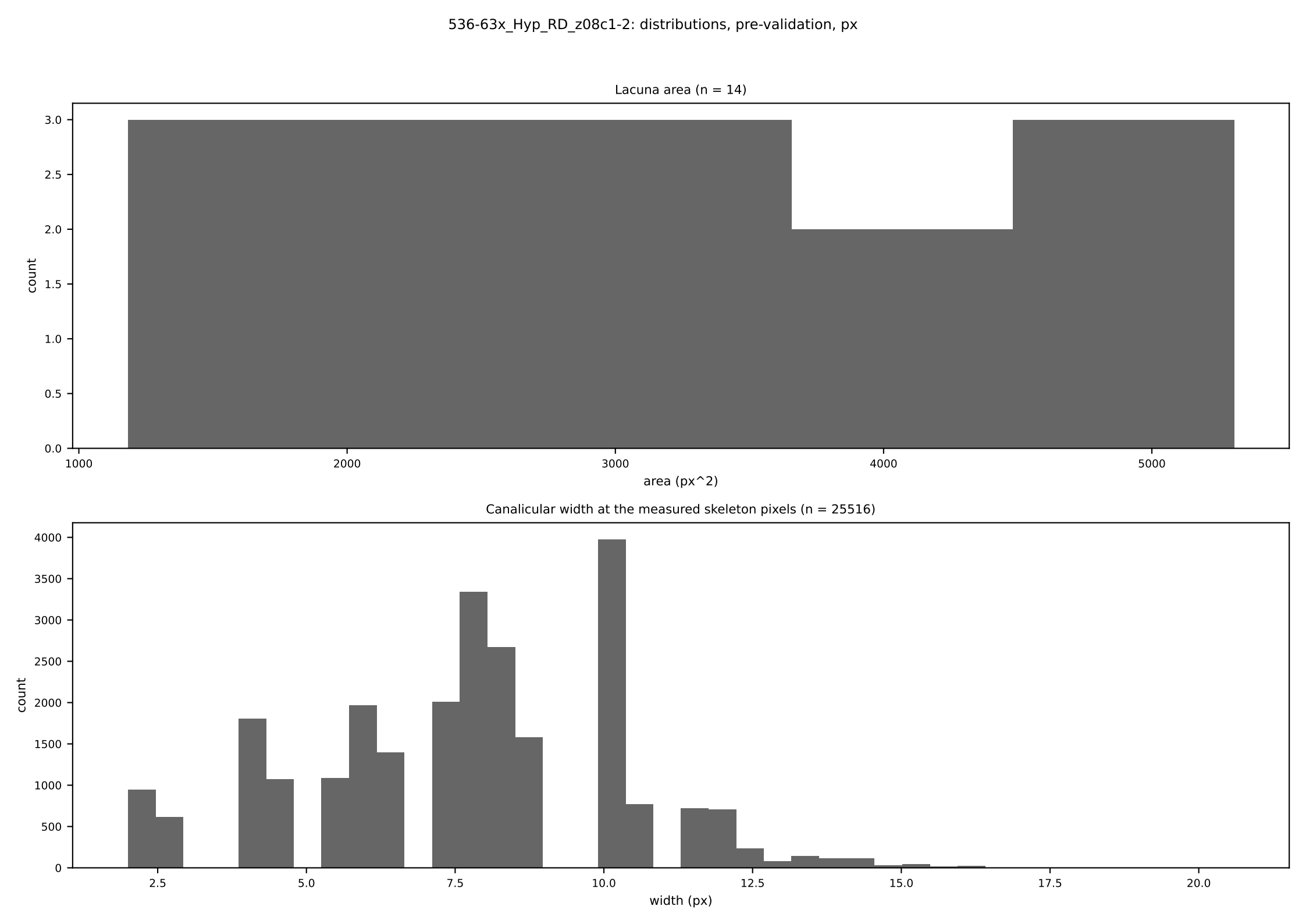
4	20	632	1263	3256.5	156	20.875	7.0906
5	12	275	586	321.593	14	22.9709	7.06
6	13	411	890	1092.82	61	17.9151	8.8041
7	10	286	601	967.637	42	23.039	8.2735
8	6	200	468	298.563	17	17.5626	7.9746
9	11	305	670	1297.54	48	27.0322	8.1069
10	10	349	860	1189.62	50	23.7924	7.3406
11	6	167	282	258.149	8	32.2687	7.3838
12	11	287	435	448.233	32	14.0073	7.1528
13	10	375	855	842.269	33	25.5233	8.5809
14	6	216	453	316.049	16	19.753	6.858

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	888.992	20368	42370	1	1	0.0365279	0.0330423	3.937
2	570.985	17484	37496	1	1	0.0399222	0.0344037	4.3784
3	514.931	14099	31464	1	1	0.0512802	0.0455441	3.6898
4	487.973	14467	32198	1	1	0.0436856	0.039226	4.0986
5	353.948	11970	29073	1	1	0.0229741	0.0201562	3.3903
6	348.433	12410	27080	1	1	0.0331185	0.0328656	3.731
7	227.38	9187	22623	1	1	0.031131	0.0265659	4.3979
8	274.735	10452	25593	1	1	0.0191351	0.0182863	2.1839
9	250.693	9821	24082	1	1	0.0310559	0.0278216	4.3878
10	298.534	10681	26849	1	1	0.0326748	0.032031	3.3497
11	217.037	8094	16986	1	1	0.0206326	0.0166019	2.7645
12	259.557	9006	18739	1	1	0.0318676	0.0232136	4.238
13	340.877	12243	29334	1	0.982	0.0306298	0.0291471	2.9336
14	234.309	8189	17896	0.9258	0.8017	0.0263769	0.0253129	2.5607

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	672	1081	844.696	1598.86	18	20	17
2	683	1245	796.34	1484.02	18	15	21
3	694	1052	813.127	1613.81	14	11	16
4	563	1136	709.985	1430.87	13	16	16
5	189	293	306.764	661.642	5	7	10
6	371	812	458.635	1009.29	8	11	13
7	243	396	326.735	682.541	12	7	10
8	149	242	228.865	539.742	6	9	7
9	305	602	350.978	779.181	8	9	8
10	261	536	373.551	951.796	8	11	11
11	127	186	191.581	325.534	5	5	7
12	270	400	332.22	500.487	7	7	7
13	329	642	423.019	955.595	10	15	12
14	145	227	245.451	518.801	4	6	5


536-63x_Hyp_RD_z08c1-2: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 25516)
count
width (px)